

Osaid Ziad Khader Alhawamdeh

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Artificial Intelligence & Data Science and hands-on experience building applied AI systems across computer vision, RAG, NLP, cybersecurity analytics, and predictive modeling. Built SentinelAI, a violence-detection system achieving 97%+ accuracy and 0.97 F1-score using EfficientNetB0 + Bi-LSTM, and developed SOC Copilot, a RAG-LLM graduation project for Security Operations Center analyst support. Currently building PharmaGuard AI, a pharmacist-centered RAG copilot prototype with safety-rule prompts and review-first workflows. Experienced with Python, TensorFlow/PyTorch, scikit-learn, FastAPI/Flask, Docker, SQL, vector search, and end-to-end AI system documentation.

CORE TECHNICAL SKILLS

AI & Machine Learning: supervised/unsupervised learning, model evaluation, feature engineering, XGBoost, GridSearchCV

Deep Learning & Computer Vision: TensorFlow, PyTorch, CNNs, EfficientNetB0, Bi-LSTM, video classification, real-time inference

NLP, RAG & LLM Systems: Retrieval-Augmented Generation, vector search, knowledge-grounded QA, LLM workflows, Hugging Face, spaCy, Weaviate, PGVector

Backend & Deployment: FastAPI, Flask, Docker, Docker Compose, REST APIs, Git/GitHub, PostgreSQL, SQL

Data & Engineering Tools: Python, Pandas, NumPy, scikit-learn, data preprocessing, analytical reporting, technical documentation

FEATURED AI PROJECTS

SentinelAI - Intelligent Violence Detection System | Deep Learning, Computer Vision | [Repository](#)

- Developed an automated video violence-detection pipeline using EfficientNetB0 + Bi-LSTM to model spatial and temporal video patterns.
- Achieved 97%+ accuracy and 0.97 F1-score; optimized training with lazy loading, mixed precision, and resource-aware model experimentation.
- Built a Flask-based inference dashboard for video analysis and operational review workflows.

SOC Copilot - AI-Powered RAG Assistant for SOC Analysts | Graduation Project, RAG, LLMs, Cybersecurity | [Repository](#)

- Developed a RAG-LLM assistant to support SOC analysts in investigating alerts, logs, CVEs, IoCs, malware reports, and threat-intelligence context.
- Built analyst-facing workflows for grounded Q&A, file/log analysis, Sigma rule conversion, IoC enrichment, and multi-event investigation support.
- Tech stack includes FastAPI, PostgreSQL/PGVector, local/API-backed LLM integrations, Docker, and a static security-operations dashboard.

PharmaGuard AI - Pharmacist-Centered AI Copilot Prototype | RAG, FastAPI, Next.js, Safety Workflows | [Repository](#)

- Built a review-first prototype for prescription intake, pharmacist correction, local RAG retrieval, knowledge-base governance, safety-rule prompts, and workflow traceability.
- Implemented as a non-clinical prototype using synthetic data and local placeholder knowledge; designed to keep pharmacist review mandatory.

Network Traffic Classification | Machine Learning, Cybersecurity

- Developed ML models to classify DNS-over-HTTPS (DoH) vs. Non-DoH traffic for encrypted traffic analysis and cybersecurity detection use cases.
- Processed and engineered features for a large-scale dataset exceeding 1.1 million records.

Car Price Predictor | Machine Learning, Regression Analysis, Streamlit | [Repository](#)

- Built an end-to-end XGBoost regression pipeline and interactive Streamlit app for used-car price prediction.
- Trained on 400K+ vehicle listings and tuned model performance from 0.856 to 0.898 using GridSearchCV.

EDUCATION & PROFESSIONAL DEVELOPMENT

B.Sc. in Artificial Intelligence & Data Science | Tafila Technical University | 2026

AI.SPIRE: Applied AI and ML Systems | LevelUp Economy & Istidama Consulting | Feb 2026 - Present

- 24-week intensive program, 500 hours, focused on Python for AI, Applied NLP, machine learning, RAG systems, and knowledge graphs.
- Built and deployed end-to-end AI systems integrating NLP pipelines, vector search, APIs, Docker, Git/GitHub, and AI system documentation.

HONORS & AWARDS

- 2nd Place - Irada Tech Entrepreneurship Hackathon.
- 3rd Place - IEEE Jordan AI Modeling Hackathon.
- Representative - Jordan at the 3rd International AI Forum, Egypt (2025).
- Ambassador - Al Hussein Award for Voluntary Work (2024-2025).

APPLIED EXPERIENCE

AI & Digital Skills Trainer | Community Organizations | 2024 - Present

- Delivered 14+ practical workshops in AI tools, digital skills, data literacy, and applied technology concepts for youth and community learners.
- Designed hands-on learning materials that translate technical AI concepts into accessible, task-oriented training sessions.

Intern | Crown Prince Foundation, Tafila Office | Oct 2024 & Jul 2025 - Sep 2025

- Led data collection and analysis for youth development programs, supporting analytical reporting and project-proposal preparation.

Academic Unit Lead - AI Team | Tafila Technical University | 2024

- Managed educational resources and mentored peers to support AI learning initiatives and technical knowledge-sharing activities.

CERTIFICATIONS

Gemini Certified Educator - Google for Education | Data Analytics and Business Intelligence Technology Pathway - HTUx | Prompt Engineering - Dubai Future Foundation | Network+ Certification - IT Networking Fundamentals